

RDK X3 course

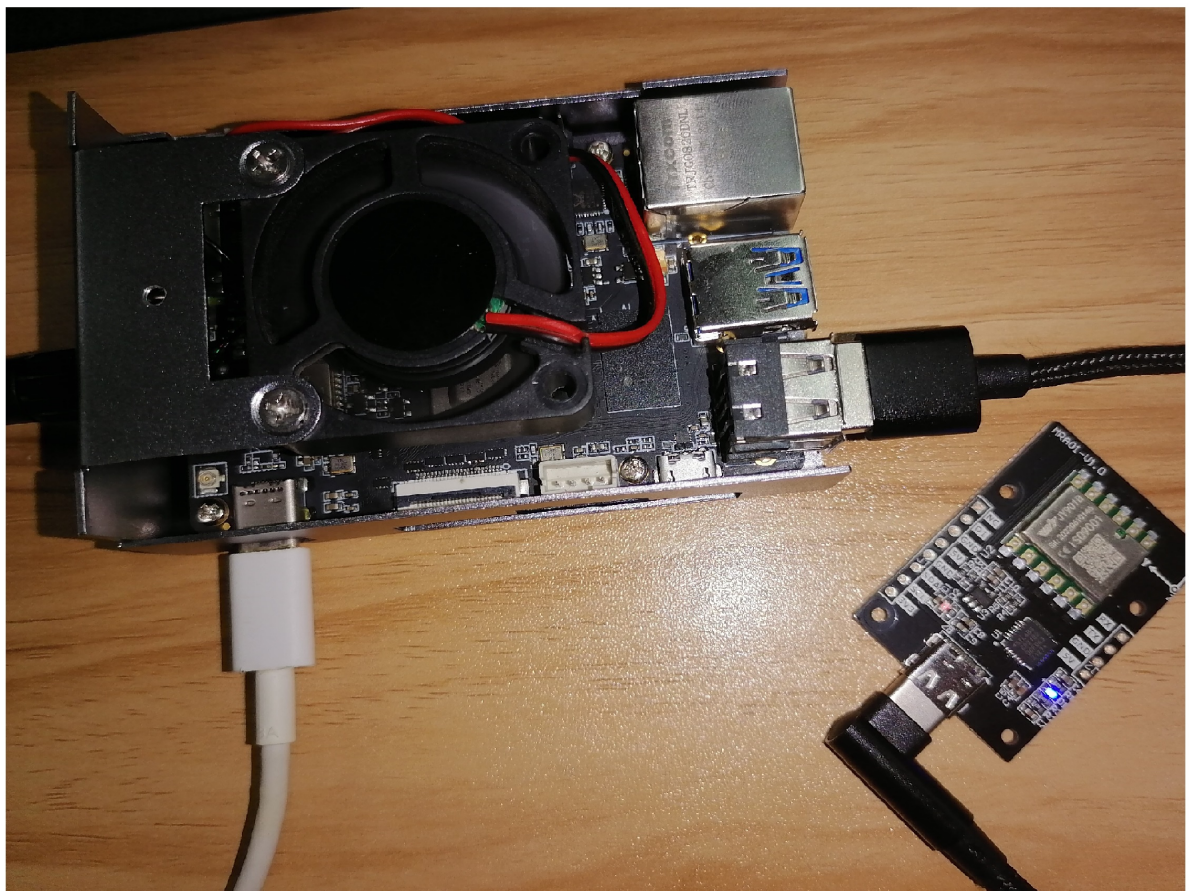
RDK X3 course

1. Hardware connection
2. Programming
3. Install serial port driver library
4. Experimental effect

Need prepared: Win10 computer, IMU module, Type-C data cable, X3 development kit.

1. Hardware connection

Connect the IMU module to the USB port of RDK X3 using a Type-C data cable.



2. Programming

About code, check code file.

If there is more than one USB device connected to the current computer, please check and confirm the USB device number of the IMU module and modify the port value.

The baud rate of USB has been set to 9600 in the program. Modify the baud value based on the baud rate of the IMU module.

```
if __name__ == '__main__':
    port = '/dev/ttyUSB0' # USB serial port
    baud = 9600 # Same baud rate as the INERTIAL navigation module
    ser = serial.Serial(port, baud, timeout=0.5)
    print("Serial is Opened:", ser.is_open)
    while(1):
        datahex = ser.read(33)
        DueData(datahex)
```

3. Install serial port driver library

If the serial port driver library is not installed in Python 3, please run the following command to install it.

```
pip3 install pyserial
```

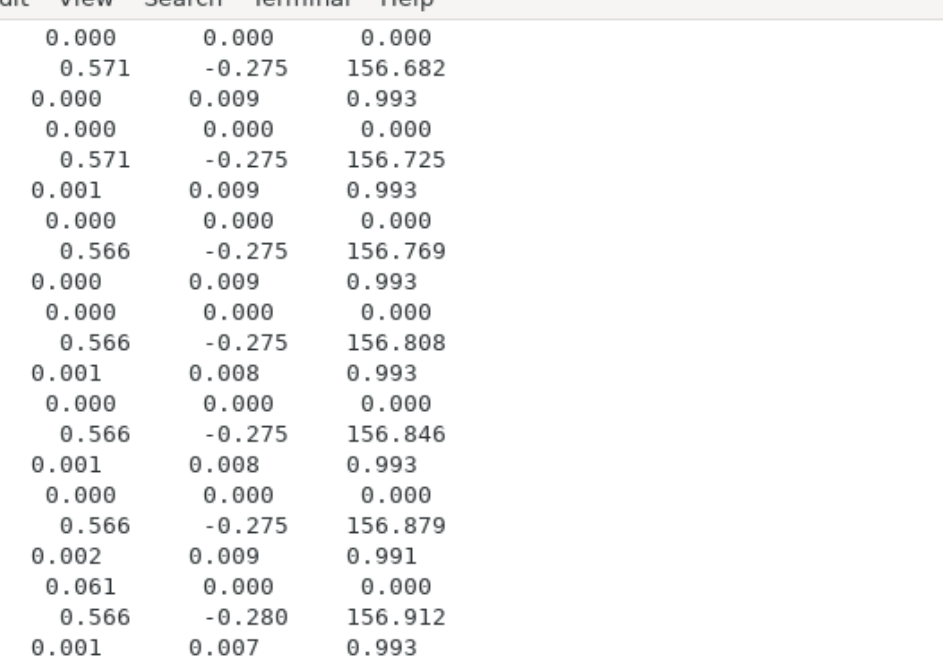
4. Experimental effect

Import the imu-usb.py file into the system, open the terminal, enter the corresponding folder, and then run the following command.

```
sudo python3 imu_usb.py
```

Entering the password again (the password is the login password, default is: **sunrise**)

You can see that the terminal keeps printing the data of the IMU module, and when the posture of the IMU module is changed, the data will change accordingly.



```
sunrise@ubuntu: ~/ROS_teeee
File Edit View Search Terminal Help
gyro: 0.000 0.000 0.000
angle: 0.571 -0.275 156.682
acc: 0.000 0.009 0.993
gyro: 0.000 0.000 0.000
angle: 0.571 -0.275 156.725
acc: 0.001 0.009 0.993
gyro: 0.000 0.000 0.000
angle: 0.566 -0.275 156.769
acc: 0.000 0.009 0.993
gyro: 0.000 0.000 0.000
angle: 0.566 -0.275 156.808
acc: 0.001 0.008 0.993
gyro: 0.000 0.000 0.000
angle: 0.566 -0.275 156.846
acc: 0.001 0.008 0.993
gyro: 0.000 0.000 0.000
angle: 0.566 -0.275 156.879
acc: 0.002 0.009 0.991
gyro: 0.061 0.000 0.000
angle: 0.566 -0.280 156.912
acc: 0.001 0.007 0.993
gyro: 0.000 0.000 0.000
angle: 0.566 -0.280 156.945
```

